

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Semester : V

Branch : Electrical and Power/Electrical & Electronics Engineering/Electrical Engineering

Subject Code & Name: Electrical Safety [BTEOE505B]

Max Marks: 60

Date: 22/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Electrical hazards include shock, electrical arcs, blasts, and or faulty equipment.						
	<table border="1"> <tr> <td>a. Broken</td> <td>b. Double-insulated</td> <td>c. Polished</td> <td>d. Secure</td> </tr> </table>	a. Broken	b. Double-insulated	c. Polished	d. Secure	CO2	1
a. Broken	b. Double-insulated	c. Polished	d. Secure				
2	 gives a leakage current somewhere to go and keeps workers away from becoming part of the circuit.						
	<table border="1"> <tr> <td>a. De-energizing</td> <td>b. Energizing</td> <td>c. Grounding</td> <td>d. Guarding</td> </tr> </table>	a. De-energizing	b. Energizing	c. Grounding	d. Guarding	CO2	1
a. De-energizing	b. Energizing	c. Grounding	d. Guarding				
3	Specially designed PPE for electrical work includes insulating gloves, matting, blankets, and covers.						
	<table border="1"> <tr> <td>a. Plastic</td> <td>b. Rubber</td> <td>c. Steel</td> <td>d. Wood</td> </tr> </table>	a. Plastic	b. Rubber	c. Steel	d. Wood	CO2	1
a. Plastic	b. Rubber	c. Steel	d. Wood				
4	The "Can't Let Go" range of current flow is typically between						
	<table border="1"> <tr> <td>a. 3-9 ma</td> <td>b. 9-25 ma</td> <td>c. 25-60 ma</td> <td>d. 1-3 ma</td> </tr> </table>	a. 3-9 ma	b. 9-25 ma	c. 25-60 ma	d. 1-3 ma	CO1	1
a. 3-9 ma	b. 9-25 ma	c. 25-60 ma	d. 1-3 ma				
5	Electrical injuries are commonly caused by electric shock.						
	<table border="1"> <tr> <td>a. Unsafe equipment .</td> <td>b. An unsafe environment</td> <td>c. Unsafe work practices.</td> <td>d. All of the above</td> </tr> </table>	a. Unsafe equipment .	b. An unsafe environment	c. Unsafe work practices.	d. All of the above	CO2	1
a. Unsafe equipment .	b. An unsafe environment	c. Unsafe work practices.	d. All of the above				
6	The effects of an electrical shock on the body depend upon all of the following except						
	<table border="1"> <tr> <td>a. Current</td> <td>b. Path</td> <td>c. Duration</td> <td>d. Body weight.</td> </tr> </table>	a. Current	b. Path	c. Duration	d. Body weight.	CO2	1
a. Current	b. Path	c. Duration	d. Body weight.				

7	 burns occur when current flows along the skin or within the body.				CO2	1
	a.Fire	b.Electrical	c.Electrical or fire	d. None of the answers		
8	You should always turn off or remove all power making alterations to an electrical circuit.				CO2	1
	a.After	b.Before or after	c.Before	d. None of the answers		
9	is a conducting object through which a direct connection to earth is established.				CO2	1
	a. Equipment-bonding	b.Equipment-grounding	c.Grounding-electrode	d. Either A or C		
10	Both voltages and body resistances will reduce the degree of electrical shock or damage to the human body.				CO2	1
	a. Lower; higher	b. Higher; lower	c.Higher; higher	d. Lower; lower		
11	Which organization in India enforces electrical safety regulations?				CO1	1
	a. Bureau of Indian Standards (BIS)	b. Ministry of Power	c. Central Electricity Authority (CEA)	d. Occupational Safety and Health Administration (OSHA)		
12	Which regulation mandates the use of Personal Protective Equipment (PPE) for electrical work?				CO1	1
	a. OSHA 1910.132	b. IEEE 1584	c. NFPA 70B	d. IEC 61850		
Q. 2	Solve the following.					12
A)	List and explain in short safety equipment used in electrical work.				CO1	6
B)	What are Causes of Electrical Hazards explain Primary and Secondary Hazards in Electrical work				CO2	6

Q.3	Solve the following.		12
A)	Define Reliability-Centered Maintenance (RCM) and explain its significance in electrical maintenance	CO2	6
B)	Discuss the role of safety considerations in RCM-based electrical maintenance programs	CO4	6
Q.4	Solve Any Two of the following.		12
A)	Which are general requirements of grounding and bonding.	CO1	6
B)	Explain Six-Step Safety Methods in Electrical Work	CO2	6
C)	What is the one-minute Electrical Safety Audit .Which are Safety Equipment and Procedures for Low, Medium, and High Voltage Systems	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Why is Safety Policy important in an Electrical Safety Program	CO2	6
B)	What are the essential first aid and rescue techniques in electrical safety	CO3	6
C)	Explain the significance of safety meetings and audits in accident prevention	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	What is the National Electrical Safety Code (NESC), and why is it important	CO2	6
B)	What are the major provisions of the Indian Electricity Act related to electrical safety	CO1	6
C)	What are the key provisions of the Occupational Safety and Health Administration (OSHA) standards related to electrical safety	CO2	6
	*** End ***		